

Qn	Suggested Answer
6	
(a)(i)	$I = \varepsilon \sigma T_{sun}^4$ $= (1)(5.67 \times 10^{-8})(5778^4)$
	$\text{Power} = I \times 4\pi r_{sun}^2$ $= \varepsilon \sigma T_{sun}^4 \times 4\pi r_{sun}^2$ $= (1)(5.67 \times 10^{-8})(5778^4) \times 4\pi (6.96 \times 10^8)^2$ $= 3.85 \times 10^{26} \text{ W (3sf)}$
(a)(ii)	$\text{intensity} = \frac{\text{power}}{\text{area over which power is distributed}}$ $S_0 = \frac{3.847 \times 10^{26}}{4\pi (1.5 \times 10^{11})^2}$
	$= 1361 \text{ W m}^{-2}$
	$= 1360 \text{ W m}^{-2} \text{ (3sf) (shown)}$
(b)	The incoming solar radiation emits mainly in the visible light wavelengths between <u>0.4 μm</u> and <u>0.7 μm</u> .
	From Fig. 6.3, the <u>absorption</u> of thermal radiation by the atmosphere <u>at these wavelengths</u> is <u>small</u> relative to that absorbed at other wavelengths (UV and Infra-red).
(c)(i)	Absorbed radiation = Emitted radiation $\sigma T_E^4 - (1 - \varepsilon_A) \sigma T_E^4 = 2\varepsilon_A \sigma T_A^4$
	$\varepsilon_A \sigma T_E^4 = 2\varepsilon_A \sigma T_A^4$
	$T_E^4 = 2T_A^4 \text{ (shown)}$
(c)(ii)	Substituting the equation in (c)(i) into the given equation for the radiative balance at the surface of the Earth, $(0.25 - 0.075)1360 + (0.8)(5.67 \times 10^{-8})\frac{1}{2}T_E^4$ $= (5.67 \times 10^{-8})T_E^4$
	$T_E = 289 \text{ K (3sf)}$
(d)	From Fig. 6.3, carbon dioxide <u>absorbs</u> in the <u>infra-red wavelengths</u> . Outgoing terrestrial radiation emits mainly in the infra-red wavelengths.
	An increase in carbon dioxide concentrations will cause the atmosphere to <u>absorb and re-emit more radiation</u> leading to global warming.

Qn	Suggested Answer
(e)(i)	
(e)(ii)	$\text{gradient} = \frac{11.0 - 0.2}{212 - 318}$ $= -0.102 \text{ km K}^{-1}$
	$\text{adiabatic lapse rate} = -\frac{1}{\text{gradient}}$ $= -\frac{1}{(-0.102)}$ $= 9.81 \text{ K km}^{-1}$ Accept 9.64 - 10.0 K km⁻¹
(e)(iii)	loss in thermal energy = work done against gravity $-mc\Delta T = mg\Delta z$
	$c = g \left(-\frac{\Delta z}{\Delta T} \right)$ $= 9.81 \left(\frac{1}{9.81 \times 10^{-3}} \right)$
	$c = 1000 \text{ J kg}^{-1} \text{ K}^{-1}$
(f)(i)	$z_0 = 5.4 - 5.6 \text{ km}$
(f)(ii)	<u>Above z_0</u>, the <u>air parcel cools faster than the surrounding air</u> while <u>below z_0</u>, the <u>surrounding air cools faster than the air parcel</u>.

Qn	Suggested Answer
	<u>Above z_0</u> , the air parcel will contract more and become <u>denser</u> than the surrounding air and <u>sinks</u> .
	<u>Below z_0</u> , the air parcel will expand more and become <u>less dense</u> than the surrounding air and <u>rises</u> .